ADRITA RAHMAN TORY

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SUMMARY

Artificial Intelligence and Cybersecurity Researcher with a B.Sc. in Computer Science and Engineering and an ongoing Professional Master's in IT. Dedicated to advancing safe AI and cutting-edge security with a human-focused, futuristic vision.

EDUCATION

Professional Master's in IT | *Jahangirnagar University, Savar, Dhaka, Bangladesh.*
Aug 2025 – Expected Sep 2026 (AQF Level 9)
***Ongoing Thesis Title**: Multi-Frequency Memory for IoT Intrusion Detection*

B.Sc. in Computer Science & Engineering | *Bangladesh University of Business & Technology, Dhaka, Bangladesh.*
Jul 2021 – Mar 2025
Final grade: 3.55/4.00 (AQF Level 7)

***Capstone Project:** AI-Powered Career Platform for University Students; Project Link: https://github.com/adritatori/bubt_internal_sns*

Diploma in Computer Engineering | *Dhaka Mohila Polytechnic Institute, Dhaka, Bangladesh.*
Jun 2016 – Mar 2021
Specialisation: Web Mastering; Final grade: 3.48/4.00 (AQF Level 5/6)
***Industry Attachment**: Internship on Android Apps Development at BD TASK LTD, Dhaka Bangladesh.*

LANGUAGES & ENGLISH PROFICIENCY

Bengali: Native
IELTS (Academic): Overall 7.0 | Reading 7.5 | Listening 8.0 | Speaking 6.5 | Writing 6.0

RESEARCH EXPERIENCE

University Research Assistant | *Bangladesh University of Business & Technology*
May 2025 – Aug 2026
Supervised by Md. Saiful Rahman (BUBT)

- Conducted focused research on AI-based cyber threat detection and adversarial AI, including comprehensive experimental evaluations
- Contributed to academic papers, preparing figures, summarising findings, and supporting writing and documentation
- Built research capacity among undergraduate students in cybersecurity at BUBT through workshops and one-on-one mentoring.

Research Assistant | *Cyber Research Hub BUBT*
Jul 2024 – Jul 2026
Supervised by Dr. Khondokar Fida Hasan (UNSW)

- Supported research on privacy-preserving 6G infrastructure, federated learning, and explainable and trustworthy AI in cybersecurity; ran experimental evaluations and assisted in building research prototypes.
- Ensured research activities complied with privacy, governance, and policy requirements across Australian and global contexts.

PUBLICATIONS

Journal Articles

Tory, A. R., & Hasan, K. F. (2025). *An Evaluation Framework for Network IDS/IPS Datasets: Leveraging MITRE ATT&CK and Industry Relevance Metrics.* Computers & Security. **Published**

Hasan, K. F., Hughes, W., Tory, A. R., Campbell, C., & Turkay, S. (2025). *Game On: A Developmental Approach to Cyber Escape Room for Cybersecurity Governance and Policy Education.* Education Sciences, MDPI. **Published**

Conference Papers

Tory, A. R., Hasan, K. F., Rahman, M. S., Koroniotis, N., & Moni, M. A. (2025). *Mind the Gap: Missing Cyber Threat Coverage in NIDS Datasets for the Energy Sector.* **Presented at:** *ICBDIML 2025*.

Hasan, K. F., Hughes, W., & Tory, A. R. (2025). *Gamifying Cyber Governance: A Virtual Escape Room to Transform Cybersecurity Policy Education.* **Presented at***: ISSOTL 2025.*

Tonmoy, T. R., Tory, A. R., Islam, M. R., Habib, M. A., & Rahaman, M. A. (2026). *Evaluating Provenance-Based Defenses Against Prompt Injection in LLM Agents: What Taint Tracking Can and Cannot See.* **Accepted at:** *IEEE CSDE 2026.*

Chakraborty, S., Tory, A. R., Hasan, K. F., & Rahman, M. A. (2026). *An Empirical Evaluation of Quantum Kernel Stability in Network Intrusion Detection.* **Accepted at:** *IEEE iCOSTE 2026.*

Tonmoy, T. R., Tory, A. R., & Hasan, K. F. (2026). *DevSecOps 2.0: Shift-Up and a Sec-Trust-Gov Model for AI-Enabled Software Development.* **Accepted at:** *IEEE ICCIT 2026.*

Rahi, S. M., Siddika, A., Maruf, I. H., Rahaman, M. A., Hasan, K. F., & Tory, A. R. (2026). *A Decade of Healthcare Cyber Threats: Empirical Analysis, Evidence-Based Prioritisation, and AI Threat Model.* **Presented at:** *ML4CS 2026.*

Tory, A. R., Ali, A. S., Rahman, M. A., & Hasan, K. F. (2026). *Privacy, Robustness, and Fairness Trade-offs in Federated Intrusion Detection: Geometric Indistinguishability at the Aggregation Interface.* **Presented at:** *ML4CS 2026.*

Mukta, J. S., Sanjida, R., Tory, A. R., Rahman, M. S., & Hasan, K. F. (2026). *Preprocessing Failure and Adversarial Detection in Depthwise-Separable Edge Vision Systems.* **Presented at:** *ML4CS 2026.*

Khanom, S., Khan, S. A., Tory, A. R., Hasan, M. A., & Hasan, K. F. (2026). *Multi-LLM Consensus Framework for Evaluating Banking-Sector NIDS Dataset Coverage of MITRE ATT&CK Techniques.* **Presented at:** *ML4CS 2026.*

PROFESSIONAL SERVICE

- Peer Reviewer, COMPAS 2025 — IEEE Computer Society Bangladesh Chapter
- Subreviewer / External Reviewer— STI 2024 and STI 2024
- Designed and facilitated research-training workshops for CSE students at BUBT, including "Cybersecurity Under the Umbrella: Paths, Reality, and Research Readiness" and "From Observation to Research Question", covering research-gap identification and thesis/publication scoping

STUDENT MENTORED

11 Undergraduate Researchers, Cyber Research Hub BUBT

Through Cyber Research Hub BUBT, I mentor undergraduate researchers on cybersecurity and AI research projects, providing guidance on research methodology, literature review, experimentation, and academic writing. The students work on topics including network security, quantum machine learning, AI security, adversarial machine learning, and trustworthy AI. I have also collaborated with mentees on three accepted international conference papers arising from their research, including studies on adversarial detection, NIDS dataset coverage using multi-LLM consensus, and healthcare cyber-threat modelling.

PROFESSIONAL EXPERIENCE

Software Engineer *(Part Time)* | *Sun Co. Ltd., Dhaka, Bangladesh*
Jan 2023 – Jul 2024

Intern, Software Engineering | *Sun Co. Ltd., Dhaka, Bangladesh*
Jul 2022 – Dec 2022

RPL/RTO Trainer | *Bangladesh Technical Education Board, Dhaka, Bangladesh*
Mar 2020 – Jun 2021

Guest Trainer, IT & Web Development | *Ideal Institute of Technology, Rishda Bangladesh, Kaizen IT LTD, Technical Teachers' Training College, New Trizon IT — Dhaka, Bangladesh*
2019 – 2022

- Delivered Course wise training sessions in IT operations, web development, and graphic design across five institutes

CERTIFICATIONS

- CompTIA Security+ — CompTIA (2025)
- Cisco Network Security: Intrusion Detection and Prevention, By: Lisa Bock — LinkedIn Learning (2026)
- RAG, AI Apps, and AI Agents for Cybersecurity and Networking, By: Pearson and Omar Santos — LinkedIn Learning (2026)
- IT Essentials — Cisco (2020)
- National Certificate-IV in CBT&A Methodology for Trainers & Assessors — BTEB (2021)

TECHNICAL SKILLS

- **AI/ML & LLM Security**
  Adversarial detection & robustness pipelines (PyTorch, TensorFlow, Scikit-learn, Keras) · Federated learning with differential privacy (Flower, PySyft, Opacus) · Explainable AI (SHAP, LIME) · Basics of RAG pipelines & AI agent architectures((LangChain, LangGraph) · Basics of LLM security in SOC environments (AI threat modelling, prompt injection & jailbreaking, and AI supply chain security)
- **Cybersecurity**
  Threat-coverage frameworks (MITRE ATT&CK, MITRE ATLAS, CVSS v3.x, OWASP Top 10, OWASP LLM Top 10) · Network & pen-testing tools (Zeek, Wireshark, Snort, Nmap, Nessus, Metasploit) · SIEM (Splunk, ELK Stack) · Governance compliance (NIST CSF, GDPR, ISO 27001, AI RMF, EU AI Act, Essential Eight)
- **Research & Statistics**
  Hypothesis testing, cross-validation, experimental design · Data analysis (Pandas, NumPy, SciPy) · Visualisation (Matplotlib, Seaborn) · Academic writing (LaTeX, Overleaf, EndNote) · 11 peer-reviewed publications (incl. 3 accepted) · Systematic literature review
- **Programming & Systems**
  Python (advanced), C++, Bash · Web (JavaScript/TypeScript/React) · Databases (MySQL, PostgreSQL, MongoDB) · Docker, Git/GitHub, Linux (Ubuntu, Kali)